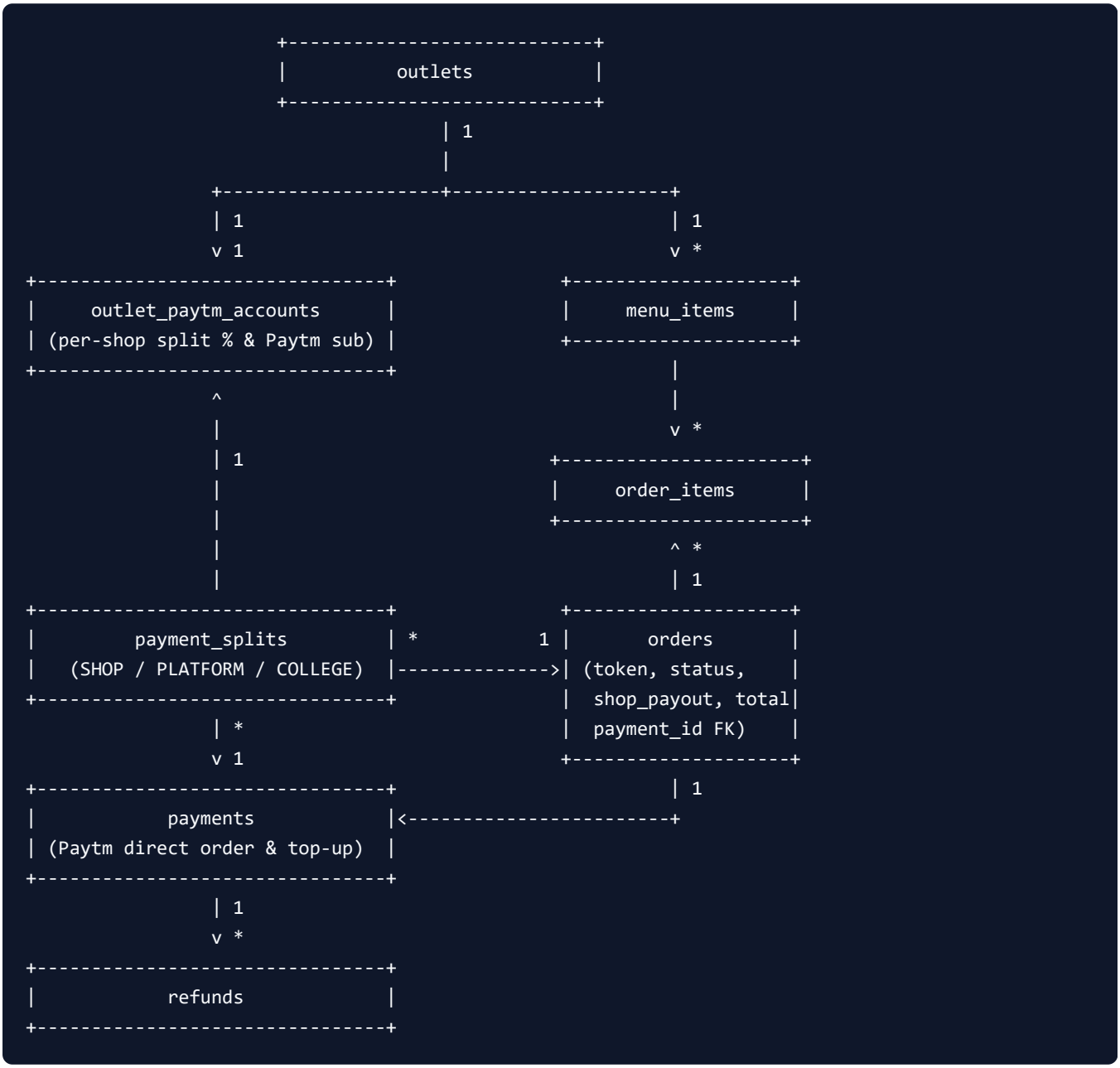
V FOODS Database Architecture Specification

This document details the PostgreSQL / Supabase database architecture for the V FOODS campus dining platform, including the complete schema, functions, triggers, security policies, and the Paytm three-way revenue settlement system.

1. Schema Overview & Entity Relationship



2. Table Definitions

2.1 Core Entities

`outlets`

Campus dining outlets and event food stalls. - `id` (text, PK): e.g. `'main-canteen'`, `'tech-cafe'`. - `name` (text, NOT NULL): Display name. - `location` (text, NOT NULL): Physical location on campus. - `is_event` (boolean, DEFAULT false): Whether outlet operates in event mode. - `is_open` (boolean, DEFAULT true): Current operating status.

`menu_items`

Food and drink items offered by outlets. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `outlet_id` (text, FK `outlets.id` ON DELETE CASCADE). - `name` (text, NOT NULL). - `price` (int, CHECK price > 0): In INR. - `available` (boolean, DEFAULT true). - `is_veg` (boolean, DEFAULT true). - `category` (text, DEFAULT 'General'). - `available_from` (time, NULL): Available start time window. - `available_to` (time, NULL): Available end time window. - `stock_qty` (int, NULL): Available units; NULL for unlimited. - `reserved_qty` (int, DEFAULT 0): Temporarily locked during payment pending.

`profiles`

User profiles extending `auth.users`. - `id` (uuid, PK FK `auth.users.id` ON DELETE CASCADE). - `full_name` (text, DEFAULT ''). - `role` (user_role ENUM: `'customer'`, `'staff'`, `'shop_admin'`, `'super_admin'`, `'admin'`). - `cust_type` (cust_type ENUM: `'student'`, `'faculty'`, `'visitor'`). - `outlet_id` (text, NULL FK `outlets.id`): Assigned stall for staff/shop admins. - `phone` (text, NULL): Used for WhatsApp/SMS notifications. - `created_at` (timestamptz, DEFAULT now()).

`wallets`

Prepaid digital campus wallet balances. - `user_id` (uuid, PK FK `profiles.id` ON DELETE CASCADE). - `balance` (int, DEFAULT 0 CHECK balance >= 0). - `updated_at` (timestamptz, DEFAULT now()).

`wallet_txns`

Audit ledger of all credits, debits, orders, and refunds. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `user_id` (uuid, FK `profiles.id` ON DELETE CASCADE). - `amount` (int): Positive for credits, negative for debits. - `kind` (text, CHECK `'topup'`, `'order'`, `'refund'`, `'admin_credit'`). - `ref` (text, UNIQUE): Idempotency key (e.g. `'paytm:TXN123'`, `'order:456'`). - `note` (text, NULL). - `created_at` (timestamptz, DEFAULT now()).

`orders`

Food orders placed by students/customers. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `user_id` (uuid, FK `profiles.id` ON DELETE RESTRICT). - `outlet_id` (text, FK `outlets.id` ON DELETE RESTRICT). -

`token` (text, NULL): 3-digit daily counter pickup code. - `status` (order_status ENUM: `'payment_pending'`, `'placed'`, `'preparing'`, `'ready'`, `'collected'`, `'cancelled'`). - `payment_method` (payment_method ENUM: `'wallet'`, `'gateway'`). - `shop_payout` (int, CHECK shop_payout > 0): Raw subtotal for items. - `total` (int, CHECK total > 0): Student bill (`ceil(shop_payout * 1.05)`). - `my_profit` (int, GENERATED ALWAYS AS `(shop_payout * 5 / 100)` STORED). - `payment_id` (bigint, NULL FK `payments.id` ON DELETE SET NULL). - `pickup_slot_id` (uuid, NULL FK `pickup_slots.id`). - `group_id` (uuid, NULL). - `is_group_payer` (boolean, DEFAULT false). - `cancel_reason` (text, NULL). - `expires_at` (timestamptz, NULL): Expiration for pending gateway orders. - `created_at` (timestamptz, DEFAULT now()). - `updated_at` (timestamptz, DEFAULT now()).

`order_items`

Line items in an order with snapshot prices. - `order_id` (bigint, FK `orders.id` ON DELETE CASCADE). - `item_id` (bigint, FK `menu_items.id` ON DELETE RESTRICT). - `name` (text, NOT NULL): Snapshot at order time. - `price` (int, NOT NULL): Snapshot at order time. - `qty` (int, NOT NULL CHECK qty > 0). - PRIMARY KEY (`order_id`, `item_id`).

2.2 Paytm Payment Architecture & Three-Way Revenue Settlement

`payments`

Unified ledger of all payment attempts across both Wallet Top-Up and Direct Order Payment. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `user_id` (uuid, FK `profiles.id` ON DELETE RESTRICT). - `order_id` (bigint, NULL FK `orders.id` ON DELETE SET NULL). - `wallet_txn_id` (bigint, NULL FK `wallet_txns.id` ON DELETE SET NULL). - `amount` (numeric, CHECK amount > 0): In INR. - `payment_purpose` (text, CHECK `'ORDER'`, `'WALLET_TOPUP'`). - `payment_method` (text, CHECK `'UPI_ID'`, `'UPI_APP'`, `'UPI_QR'`, `'DEBIT_CARD'`, `'CREDIT_CARD'`, `'WALLET'`). - `status` (text, DEFAULT 'PENDING' CHECK `'PENDING'`, `'SUCCESS'`, `'FAILED'`, `'CANCELLED'`, `'REFUNDED'`). - `paytm_order_id` (text, NULL): Merchant order reference sent to Paytm. - `paytm_txn_id` (text, NULL UNIQUE): Paytm bank transaction ID (idempotency key). - `payment_reference` (text, NULL): Internal tracking reference. - `paid_at` (timestamptz, NULL): Timestamp when marked SUCCESS. - `failure_reason` (text, NULL): Paytm gateway error response message. - `created_at` (timestamptz, DEFAULT now()). - `updated_at` (timestamptz, DEFAULT now()).

`outlet_paytm_accounts`

Per-shop Paytm recipient sub-accounts with outlet-configurable split shares. - `outlet_id` (text, PK FK `outlets.id` ON DELETE CASCADE). - `paytm_account_id` (text, NOT NULL): Shop's registered Paytm merchant ID. - `onboarding_status` (text, DEFAULT 'PENDING' CHECK `'PENDING'`, `'ACTIVE'`, `'REJECTED'`). - `settlement_enabled` (boolean, DEFAULT false). - `shop_split_percentage` (numeric, DEFAULT 90 CHECK BETWEEN 0 AND 100). - `created_at` (timestamptz, DEFAULT now()). - `updated_at` (timestamptz, DEFAULT now()).

`platform_settlement_account`

Startup platform settlement account (singleton row `id = 1`). - `id` (int, PK DEFAULT 1 CHECK id = 1). - `paytm_account_id` (text, NOT NULL): V Foods company Paytm merchant ID. - `settlement_enabled` (boolean, DEFAULT false). - `platform_split_percentage` (numeric, DEFAULT 5 CHECK BETWEEN 0 AND 100). - `created_at` (timestamptz, DEFAULT now()). - `updated_at` (timestamptz, DEFAULT now()).

`college_settlement_accounts`

Campus institution settlement accounts (one active row per campus). - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `campus_name` (text, NOT NULL): e.g. `'VIT Chennai'`. - `paytm_account_id` (text, NOT NULL): University institution Paytm ID. - `onboarding_status` (text, DEFAULT 'PENDING' CHECK `'PENDING'`, `'ACTIVE'`, `'REJECTED'`). - `settlement_enabled` (boolean, DEFAULT false). - `is_active` (boolean, DEFAULT true): Only one active campus at a time. - `college_split_percentage` (numeric, DEFAULT 5 CHECK BETWEEN 0 AND 100). - `created_at` (timestamptz, DEFAULT now()). - `updated_at` (timestamptz, DEFAULT now()). - Constraint: `CREATE UNIQUE INDEX idx_college_single_active ON college_settlement_accounts(is_active) WHERE is_active = true;`

`payment_splits`

Individual settlement payouts computed for every completed order. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `payment_id` (bigint, FK `payments.id` ON DELETE CASCADE). - `recipient_type` (text, CHECK `'SHOP'`, `'PLATFORM'`, `'COLLEGE'`). - `outlet_paytm_account_id` (text, NULL FK `outlet_paytm_accounts.outlet_id` ON DELETE RESTRICT). - `platform_account_id` (int, NULL FK `platform_settlement_account.id` ON DELETE RESTRICT). - `college_account_id` (bigint, NULL FK `college_settlement_accounts.id` ON DELETE RESTRICT). - `split_amount` (numeric, CHECK split_amount >= 0). - `split_percentage` (numeric, NULL CHECK BETWEEN 0 AND 100). - `settlement_status` (text, DEFAULT 'PENDING' CHECK `'PENDING'`, `'SETTLED'`, `'FAILED'`). - `settlement_reference` (text, NULL): Paytm disbursement transaction reference. - `settled_at` (timestamptz, NULL). - `created_at` (timestamptz, DEFAULT now()). - Constraint: `sql CHECK ((recipient_type = 'SHOP' AND outlet_paytm_account_id IS NOT NULL AND platform_account_id IS NULL AND college_account_id IS NULL) OR (recipient_type = 'PLATFORM' AND platform_account_id IS NOT NULL AND outlet_paytm_account_id IS NULL AND college_account_id IS NULL) OR (recipient_type = 'COLLEGE' AND college_account_id IS NOT NULL AND outlet_paytm_account_id IS NULL AND platform_account_id IS NULL))`

`refunds`

Audit log of processed and pending customer refunds. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `payment_id` (bigint, FK `payments.id` ON DELETE RESTRICT). - `order_id` (bigint, NULL FK `orders.id` ON DELETE SET NULL). - `wallet_txn_id` (bigint, NULL FK `wallet_txns.id` ON DELETE SET NULL). - `refund_type` (text, CHECK `'ORDER_REFUND'`, `'WALLET_TOPUP_REFUND'`, `'SPLIT_REFUND'`). - `amount` (numeric, CHECK amount > 0). - `status` (text, DEFAULT 'PENDING' CHECK `'PENDING'`, `'SUCCESS'`, `'FAILED'`). - `paytm_refund_id` (text, NULL). - `reason` (text, NULL). - `created_at` (timestamptz, DEFAULT now()). - `processed_at` (timestamptz, NULL).

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3. General Payment-Confirmation Flow

1. **Frontend Isolation**: The frontend client never decides or confirms a payment succeeded. It only displays a temporary "Processing" state and listens for realtime updates from Supabase.
2. **Signed Webhook Ingestion**: When a transaction concludes, Paytm sends an HTTPS POST webhook containing `CHECKSUMHASH` and `TXNID` to the FastAPI backend at `/webhooks/paytm`.
3. **Cryptographic Checksum Verification**: The FastAPI backend recomputes the HMAC-SHA256 signature using `PAYTM_MERCHANT_KEY`. Any unsigned, forged, or altered request is rejected with HTTP 400.
4. **Idempotent DB Procedure**: The backend invokes `verify_and_record_payment()` in PostgreSQL. If the transaction was already processed, it returns immediately without duplicating work.
5. **Path Execution**:
6. For `WALLET_TOPUP`: Credits the user's wallet via `credit_wallet()` and links `wallet_txns.id`. Top-ups are never split.
7. For `ORDER`: Marks the order `placed`, issues the 3-digit token, decrements stock, and calls `calculate_order_split()` to record the 3 split rows.
8. **Network Reliability**: Because confirmation occurs server-to-server, if a student closes their browser or their battery dies right after paying in UPI, confirmation is unaffected.

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4. The Three-Way Revenue Split Explained

- **Shop Split (Configurable per Outlet)**:
- Each stall owner negotiates a split percentage (default 90%). Stored in `outlet_paytm_accounts.shop_split_percentage`.
- Differing stalls can have differing rates (e.g. 90% for Main Canteen, 85% for Tech Café).
- **Platform Split (Fixed App-Wide)**:
- V Foods platform fee (default 5%) stored in `platform_settlement_account.platform_split_percentage`.
- **College Split (Fixed per Campus)**:
- University institutional royalty (default 5%) stored in `college_settlement_accounts.college_split_percentage`.
- **Trigger Invariant**: A PostgreSQL trigger enforces that for any outlet: $$\text{Shop \%} + \text{Platform \%} + \text{College \%} = 100\%$$
- **Penny Conservation**: The college share takes the residual difference $(Total - Shop - Platform)$ to guarantee exact zero-penny rounding leakage.

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5. Security & Row Level Security (RLS)

- **No Card / PIN Data**: No column stores card PAN, CVV, expiry dates, or UPI PINs.
- **RLS Access Rules**:

- `payments` : Customers view only their own payments. Staff and Shop Admins view payments for their own outlet via the linked order. Super Admins view all.
- `payment_splits` : Shop Staff/Admins can view ONLY their own outlet's `SHOP` -type rows. `PLATFORM` and `COLLEGE` rows are visible only to Super Admins. Students have zero access.
- `outlet_paytm_accounts` : Shop Admins view and manage only their outlet's row. Super Admins manage all.
- `platform_settlement_account` & `college_settlement_accounts` : Restricted exclusively to Super Admins.
- `refunds` : Users view own refunds; staff view outlet-scoped order refunds; Super Admins view all.
